

Normalized Tables

Storage_areas

<u>StorageAreaID</u>	Length	Width	Height	Location	JobsiteID	Total Stored	Is_Container
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Materials

<u>MaterialID</u>	Name	Material Type	Length	Width	Height	Supplier Name	Total Available	Lost Amounts
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Stored_in

<u>StorageAreaID</u>	<u>MaterialID</u>	Amount
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Jobsites

<u>JobsiteID</u>	JobsiteName
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Employees

<u>EmpID</u>	Fname	Lname	Title	SupervisorID	JobsiteID
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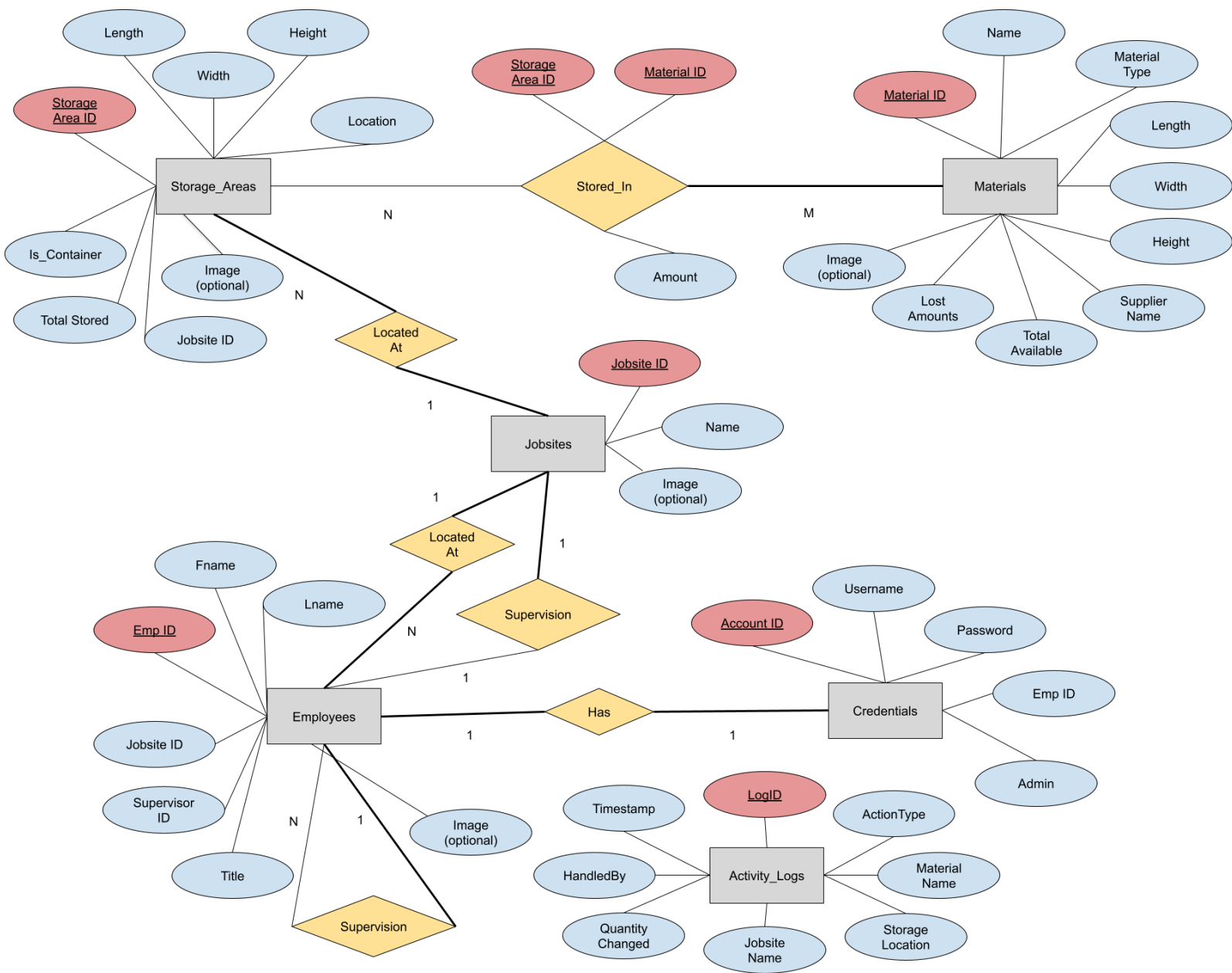
Credentials

<u>AccountID</u>	Username	Password	EmpID	Admin
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Activity_logs

<u>LogID</u>	ActionType	Material Name	Storage Location	Jobsite Name	Quantity Changed	Handled By	Timestamp
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Entity Relationship Diagram



Creation

```
CREATE TABLE Jobsites (  
    JobsiteID INT PRIMARY KEY AUTO_INCREMENT,  
    JobsiteName VARCHAR(64) NOT NULL  
);
```

```
CREATE TABLE Materials (  
    MaterialID INT PRIMARY KEY AUTO_INCREMENT,  
    Name VARCHAR(64) NOT NULL,  
    MaterialType VARCHAR(64) NOT NULL,  
    Length DECIMAL(5,2),  
    Width DECIMAL(5,2),  
    Height DECIMAL(5,2),  
    SupplierName VARCHAR(64) NOT NULL,  
    TotalAvailable SMALLINT DEFAULT 0,  
    LostAmounts SMALLINT DEFAULT 0  
);
```

```
CREATE TABLE Storage_Areas (  
    StorageAreaID INT PRIMARY KEY AUTO_INCREMENT,  
    Length DECIMAL(5,2),  
    Width DECIMAL(5,2),  
    Height DECIMAL(5,2),  
    Location VARCHAR(64) NOT NULL,  
    JobsiteID INT,  
    TotalStored SMALLINT DEFAULT 0,  
    Is_Container BOOLEAN DEFAULT FALSE,  
    FOREIGN KEY(JobsiteID) REFERENCES Jobsites(JobsiteID) ON DELETE SET NULL  
);
```

```
CREATE TABLE Employees (  
    EmpID INT PRIMARY KEY AUTO_INCREMENT,  
    Fname VARCHAR(32) NOT NULL,  
    Lname VARCHAR(32) NOT NULL,  
    Title VARCHAR(32),  
    SupervisorID INT,  
    JobsiteID INT,  
    FOREIGN KEY(JobsiteID) REFERENCES Jobsites(JobsiteID) ON DELETE SET NULL,  
    FOREIGN KEY(SupervisorID) REFERENCES Employees(EmpID) ON DELETE SET NULL  
);
```

```

CREATE TABLE Credentials (
    AccountID INT PRIMARY KEY AUTO_INCREMENT,
    Username VARCHAR(64) NOT NULL UNIQUE,
    Password VARCHAR(256) NOT NULL,
    EmpID INT NOT NULL UNIQUE,
    Admin BOOLEAN DEFAULT FALSE,
    FOREIGN KEY(EmpID) REFERENCES Employees(EmpID) ON DELETE CASCADE
);

CREATE TABLE stored_in (
    StorageAreaID INT NOT NULL,
    MaterialID INT NOT NULL,
    Amount SMALLINT NOT NULL DEFAULT 0,
    PRIMARY KEY (StorageAreaID, MaterialID),
    FOREIGN KEY(StorageAreaID) REFERENCES Storage_Areas(StorageAreaID) ON DELETE CASCADE,
    FOREIGN KEY(MaterialID) REFERENCES Materials(MaterialID) ON DELETE CASCADE
);

CREATE TABLE activity_logs (
    LogID INT AUTO_INCREMENT PRIMARY KEY,
    ActionType VARCHAR(64),
    MaterialName VARCHAR(75),
    StorageLocation VARCHAR(75),
    JobsiteName VARCHAR(75),
    QuantityChanged INT,
    HandledBy VARCHAR(75),
    Timestamp DATETIME DEFAULT CURRENT_TIMESTAMP
);

```

Images Note: Optional images are handled via **VARCHAR(2083) NULL** columns on the parent tables. These rows store absolute URL string endpoints targeting external assets rather than hosting raw image binary data directly inside the relational database. See appendix for the queries that add image columns. Activity Logs Note: the **VARCHAR(75)** columns are for allowing this format “columnName-primaryKey” (i.e, “Abc Refinery-123”) under this version of the DDD.

Indexes Note: Performance indexes are managed separately in the backend repository where they can adhere to backend designs.

Sample Entries For Database

Employees

EmpID	Fname	Lname	Title	SupervisorID	JobsiteID
1	George	Washington	Owner	1	NULL
2	John	Adams	Site Supervisor	1	1
3	Thomas	Jefferson	Site Supervisor	1	2
4	James	Madison	Foreman	2	1
5	James	Monroe	Insulator	4	1
6	John Q.	Adams	Insulator	4	1
7	Andrew	Jackson	Foreman	3	2
8	Martin	Van Buren	Insulator	7	2
9	William H.	Harrison	Insulator	7	2
10	John	Tyler	Insulator	7	2
11	James K.	Polk	Helper	4	1
12	Zachary	Taylor	Helper	7	2
13	Millard	Fillmore	Insulator	4	1

INSERT INTO Employees(Fname, LName, Title, SupervisorID, JobsiteID) VALUES("George", "Washington", "Owner", 1, NULL);

INSERT INTO Employees(Fname, LName, Title, SupervisorID, JobsiteID) VALUES("John", "Adams", "Site Supervisor", 1, 1);

INSERT INTO Employees(Fname, LName, Title, SupervisorID, JobsiteID) VALUES("Thomas", "Jefferson", "Site Supervisor", 1, 2);

INSERT INTO Employees(Fname, LName, Title, SupervisorID, JobsiteID) VALUES("James", "Madison", "Foreman", 2, 1);

INSERT INTO Employees(Fname, LName, Title, SupervisorID, JobsiteID) VALUES("James", "Monroe", "Insulator", 4, 1);

INSERT INTO Employees(Fname, LName, Title, SupervisorID, JobsiteID) VALUES("John Q.", "Adams", "Insulator", 4, 1);

```

INSERT INTO Employees(Fname, LName, Title, SupervisorID, JobsiteID) VALUES("Andrew",
"Jackson", "Foreman", 3, 2);
INSERT INTO Employees(Fname, LName, Title, SupervisorID, JobsiteID) VALUES("Martin",
"Van Buren", "Insulator", 7, 2);
INSERT INTO Employees(Fname, LName, Title, SupervisorID, JobsiteID) VALUES("William H.",
"Harrison", "Insulator", 7, 2);
INSERT INTO Employees(Fname, LName, Title, SupervisorID, JobsiteID) VALUES("John",
"Tyler", "Insulator", 7, 2);
INSERT INTO Employees(Fname, LName, Title, SupervisorID, JobsiteID) VALUES("James K.",
"Polk", "Helper", 4, 1);
INSERT INTO Employees(Fname, LName, Title, SupervisorID, JobsiteID) VALUES("Zachary",
"Taylor", "Helper", 7, 2);
INSERT INTO Employees(Fname, LName, Title, SupervisorID, JobsiteID) VALUES("Millard",
"Fillmore", "Insulator", 4, 1);

```

Storage Areas*

<u>StorageA</u> <u>realID</u>	Length	Width	Height	Location	<u>JobsiteID</u>	TotalStor ed	Is_Contai ner
1	30.48	30.48	7.62	Laydown 1	1	10	False
2	15.24	15.24	7.62	Laydown 2	2	20	False
3	12.2	2.4	2.6	Laydown 1	1	20	True
4	12.2	2.4	2.6	Laydown 1	1	25	True

```

INSERT INTO Storage_Areas(Length, Width, Height, Location, JobsiteID, TotalStored,
Is_Container) VALUES (30.48, 30.48, 7.62, "Laydown 1", 1, 10, False);
INSERT INTO Storage_Areas(Length, Width, Height, Location, JobsiteID, TotalStored,
Is_Container) VALUES (15.24, 15.24, 7.62, "Laydown 2", 2, 20, False);
INSERT INTO Storage_Areas(Length, Width, Height, Location, JobsiteID, TotalStored,
Is_Container) VALUES (12.2, 2.4, 2.6, "Laydown 1", 1, 20, True);
INSERT INTO Storage_Areas(Length, Width, Height, Location, JobsiteID, TotalStored,
Is_Container) VALUES (12.2, 2.4, 2.6, "Laydown 1", 1, 25, True);

```

Materials*

<u>Material ID</u>	Name	Material Type	Length	Width	Height	Supplier Name	Total Available	Lost Amounts
1	"Insulation 1"	"Insulation"	0.92	0.92	0.92	"Brand 1"	10	0
2	"Insulation 2"	"Insulation"	0.92	0.92	0.61	"Brand 1"	30	0
3	"Insulation 3"	"Insulation"	1.22	1.22	1.22	"Brand 2"	25	10
4	"Tape 1"	"Tape"	0.3	0.3	0.3	"Brand 3"	10	0

INSERT INTO Materials(Name, MaterialType, Length, Width, Height, SupplierName, TotalAvailable, LostAmounts) VALUES("Insulation 1", "Insulation", 0.92, 0.92, 0.92, "Brand 1", 10, 0);

INSERT INTO Materials(Name, MaterialType, Length, Width, Height, SupplierName, TotalAvailable, LostAmounts) VALUES("Insulation 2", "Insulation", 0.92, 0.92, 0.61, "Brand 1", 30, 0);

INSERT INTO Materials(Name, MaterialType, Length, Width, Height, SupplierName, TotalAvailable, LostAmounts) VALUES("Insulation 3", "Insulation", 1.22, 1.22, 1.22, "Brand 2", 25, 10);

INSERT INTO Materials(Name, MaterialType, Length, Width, Height, SupplierName, TotalAvailable, LostAmounts) VALUES("Tape 1", "Tape", 0.3, 0.3, 0.3, "Brand 3", 10, 0);

Jobsites

<u>JobsiteID</u>	JobsiteName
1	ABC Data Center
2	XYZ Refinery

INSERT INTO Jobsites(JobsiteName) VALUES("ABC Data Center");

INSERT INTO Jobsites(JobsiteName) VALUES("XYZ Refinery");

Stored_In

<u>StorageAreaID</u>	<u>MaterialID</u>	Amount
1	4	10
2	2	20
3	1	10
3	3	10
4	3	15
4	2	10

```

INSERT INTO Stored_In(StorageAreaID, MaterialID, Amount) Values(1, 4, 10);
INSERT INTO Stored_In(StorageAreaID, MaterialID, Amount) Values(2, 2, 20);
INSERT INTO Stored_In(StorageAreaID, MaterialID, Amount) Values(3, 1, 10);
INSERT INTO Stored_In(StorageAreaID, MaterialID, Amount) Values(3, 3, 10);
INSERT INTO Stored_In(StorageAreaID, MaterialID, Amount) Values(4, 3, 15);
INSERT INTO Stored_In(StorageAreaID, MaterialID, Amount) Values(4, 2, 10);

```

Credentials

<u>AccountID</u>	Username	Password	<u>EmpID</u>	Admin
1	GWashington	April1789	1	TRUE
2	JAdams	March1797	2	TRUE
3	TJefferson	March1801	3	TRUE
4	JMadison	March1809	4	FALSE
5	JMonroe	March1817	5	FALSE
6	JQAdams	March1825	6	FALSE
7	AJackson	March1829	7	FALSE
8	MVanBuren	March1837	8	FALSE
9	WHHarrison	March1841	9	FALSE
10	JTyler	April1841	10	FALSE
11	JKPolk	March1845	11	FALSE

12	ZTaylor	March1849	12	FALSE
13	MFillmore	July1850	13	FALSE

```

INSERT INTO Credentials(Username, Password, EmpID, Admin) VALUES("GWashington",
"April1789", 1, TRUE);
INSERT INTO Credentials(Username, Password, EmpID, Admin) VALUES("JAdams",
"March1797", 2, TRUE);
INSERT INTO Credentials(Username, Password, EmpID, Admin) VALUES("TJefferson",
"March1801", 3, TRUE);
INSERT INTO Credentials(Username, Password, EmpID) VALUES("JMadison", "March1809",
4);
INSERT INTO Credentials(Username, Password, EmpID) VALUES("JMonroe", "March1817", 5);
INSERT INTO Credentials(Username, Password, EmpID) VALUES("JQAdams", "March1825",
6);
INSERT INTO Credentials(Username, Password, EmpID) VALUES("AJackson", "March1829",
7);
INSERT INTO Credentials(Username, Password, EmpID) VALUES("MVanBuren", "March1837",
8);
INSERT INTO Credentials(Username, Password, EmpID) VALUES("WHHarrison",
"March1841", 9);
INSERT INTO Credentials(Username, Password, EmpID) VALUES("JTylor", "March1845", 10);
INSERT INTO Credentials(Username, Password, EmpID) VALUES("JKPolk", "March1845", 11);
INSERT INTO Credentials(Username, Password, EmpID) VALUES("ZTaylor", "March1849", 12);
INSERT INTO Credentials(Username, Password, EmpID) VALUES("MFillmore", "July1850",
13);

```

Password Note: These passwords are plain text only for demo purposes.

*Length, width, and height are all in meters in these examples. Conversions to other measurements should be handled on either the frontend or backend.

Useful Queries

Get Leadership:

```
SELECT DISTINCT emp1.empid AS EmpID, CONCAT(emp1.fname, ' ', emp1.lname) AS Name,  
               emp1.title AS Title  
FROM employees as emp1  
JOIN employees as emp2 ON emp2.supervisorid = emp1.empid  
LIMIT <num> OFFSET <num>;
```

Get Subordinates:

```
SELECT * FROM employees WHERE SupervisorID = <num>;
```

Get Employees At Jobsites:

```
SELECT employees.empid AS EmpID, employees.fname AS FirstName, employees.lname AS  
       LastName, employees.title AS Title,  
       jobsites.jobsiteid AS Jobsite  
FROM employees  
LEFT JOIN jobsites ON employees.jobsiteid = jobsites.jobsiteid  
LIMIT <num> OFFSET <num>;
```

Unassigned Employees:

```
SELECT EmpID, Fname AS FirstName, Lname AS LastName, Title  
FROM employees  
WHERE JobsiteID IS NULL;
```

Get Material Amounts In Locations At Jobsites:

```
`SELECT materials.name AS Name, materials.suppliername AS Brand,  
       storage_areas.location AS Location,  
       jobsites.jobsiteid AS Jobsite,  
       stored_in.amount AS Amount  
FROM materials  
JOIN stored_in ON materials.materialid = stored_in.materialid  
JOIN storage_areas ON storage_areas.storageareaid = stored_in.storageareaid  
JOIN jobsites ON jobsites.jobsiteid = storage_areas.jobsiteid  
ORDER BY materials.name  
LIMIT <num> OFFSET <num>;`
```

Get The Materials & Sort By The Lowest Available Amounts:

```
SELECT * FROM materials ORDER BY totalavailable ASC LIMIT <num>;
```

Empty Storage Areas:

```
SELECT sa.StorageAreaID, sa.Location, j.JobsiteName
FROM storage_areas AS sa
LEFT JOIN jobsites as j ON sa.JobsiteID = j.JobsiteID
LEFT JOIN stored_in as si on sa.StorageAreaID = si.StorageAreaID
WHERE si.StorageAreaID IS NULL;
```

Inventory Discrepancy Audit:

```
SELECT
    m.MaterialID AS MaterialID,
    m.Name AS MaterialName,
    m.TotalAvailable AS CatalogTotal,
    IFNULL(SUM(si.Amount), 0) AS PhysicalTotal,
    (m.TotalAvailable - IFNULL(SUM(si.Amount), 0)) AS Discrepancy
FROM materials AS m
LEFT JOIN stored_in AS si ON m.MaterialID = si.MaterialID
GROUP BY m.MaterialID, m.Name, m.TotalAvailable
HAVING Discrepancy <> 0;
```

Appendix

For adding an Image link column:

```
ALTER TABLE jobsites ADD COLUMN Image VARCHAR(2083) NULL DEFAULT NULL;
```

```
ALTER TABLE materials ADD COLUMN Image VARCHAR(2083) NULL DEFAULT NULL;
```

```
ALTER TABLE storage_areas ADD COLUMN Image VARCHAR(2083) NULL DEFAULT NULL;
```

```
ALTER TABLE employees ADD COLUMN Image VARCHAR(2083) NULL DEFAULT NULL;
```